

# M2 COMMUNITY NODE

## Field Operations Runbook

v1.0 | April 2026

*Field-deployable decentralized communications — ATAK · Matrix · Mesh · Monero*

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## Power-On Sequence

1. UPS (BC600R) — press power, confirm green LED.
2. GL.iNet router — power on, wait 60 seconds.
3. TP-Link switch — power on.
4. Pi #1 (comms, 192.168.8.10) — power on.
5. Pi #2 (tactical, 192.168.8.20) — power on.
6. Wait 90 seconds for services to start.
7. Kiosk touchscreen — power on last.

Total boot time: ~3 minutes. Do not test connectivity during this window.

### POST — Quick Check (30 seconds)

| URL                      | Expected            | Service        |
|--------------------------|---------------------|----------------|
| http://192.168.8.1       | GL.iNet admin panel | Router         |
| http://192.168.8.10:8081 | Community info page | Kiosk web      |
| http://192.168.8.10:8080 | Element Web login   | Matrix chat    |
| http://192.168.8.20:8080 | OTS web map         | OpenTAK Server |

If all four load: node is GO.

### Full POST (SSH)

```
ssh ps@192.168.8.10
docker ps --format "table {{.Names}}\t{{.Status}}"
# All containers should show "Up"
```

```
ssh ps@192.168.8.20
systemctl is-active opentakserver cot_parser
docker ps --format "table {{.Names}}\t{{.Status}}"
# Expected: active (twice), then all containers Up
```

## Pre-Event Checklist

Complete 30 minutes before go-live.

|                          |                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | All POST checks pass (all four URLs load on test device).                                                                               |
| <input type="checkbox"/> | Update node GPS marker — OTS Web UI → map → right-click node location → set marker. Location may have changed from build to deployment. |
| <input type="checkbox"/> | Node GPS marker visible and correct on OTS web map.                                                                                     |

|                          |                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Test phone connects to CommunityNode WiFi.                                          |
| <input type="checkbox"/> | Test phone ATAK connects and enrolls (192.168.8.20:8089 SSL).                       |
| <input type="checkbox"/> | Test phone icon appears on OTS web map with correct callsign.                       |
| <input type="checkbox"/> | GeoChat test: send a message, confirm it appears in OTS.                            |
| <input type="checkbox"/> | Reporting Strategy set to Consistent on test phone.                                 |
| <input type="checkbox"/> | Mumla voice test: 192.168.8.20 port 64738 connects, voice audible.                  |
| <input type="checkbox"/> | Element Web accessible at http://192.168.8.10:8080.                                 |
| <input type="checkbox"/> | Field cards printed and available (or QR codes on kiosk).                           |
| <input type="checkbox"/> | WiFi password posted or available for incoming team members.                        |
| <input type="checkbox"/> | Cloudflare Tunnel active — verify tak.capableenough.org loads from external device. |
| <input type="checkbox"/> | Path B: Pre-add expected remote viewer emails to Cloudflare Access approved list.   |
| <input type="checkbox"/> | If SAR: ExCheck checklist template built and ready to distribute.                   |

## Operator Onboarding — Three Paths

[!] OPERATOR / ADMIN REQUIREMENT: All administrative tasks in this section require a dedicated operations PC, laptop, or tablet — not a smartphone. The node admin must have their computer available and connected to CommunityNode WiFi (for OTS Web UI and SSH) or the internet (for Cloudflare dashboard and remote management). Ensure your admin machine is ready before onboarding begins.

```

+-----+
| New team member |
| needs map access |
+-----+
|
+-----+
| Are they physically here? |
+-----+
+---yes---no---+
|               |
+-----+ +-----+
| PATH A      | | Do they need full ATAK |
| Local WiFi  | | (CoT, GeoChat, PLI)?   |
| Full ATAK   | | +-----+
+-----+ +---no---yes---+
|               |
+-----+ +-----+
| PATH B      | | PATH C      |
| Clearnet    | | VPN + ATAK   |
| Browser     | | Operator only |
+-----+ +-----+

```

## Path Comparison

|                        | Path A: Local WiFi    | Path B: Clearnet Browser   | Path C: VPN + ATAK              |
|------------------------|-----------------------|----------------------------|---------------------------------|
| <b>Location</b>        | At the node           | Anywhere with internet     | Anywhere with internet          |
| <b>App required</b>    | ATAK-CIV              | Browser only               | Tailscale + ATAK-CIV            |
| <b>Appears on map</b>  | Yes                   | No                         | Yes                             |
| <b>GeoChat</b>         | Yes                   | No                         | Yes                             |
| <b>Voice (Mumble)</b>  | Yes                   | No                         | Yes (with Mumla)                |
| <b>Onboard time</b>    | 2–5 min               | Under 60 sec               | 5–10 min                        |
| <b>Operator effort</b> | Hand them a QR code   | Add email to approved list | Generate key, send instructions |
| <b>Revoke access</b>   | They leave WiFi range | Remove email from list     | Delete node from Headscale      |

## Path A — Local WiFi (Primary)

Who: Anyone physically at the node — SAR volunteers, community members, field teams.

What they get: Full ATAK — live position sharing, GeoChat, markers, missions, voice (Mumble), mesh relay (Meshtastic).

## Operator Walkthrough

1. Get the user on the WiFi. Tell them to connect to CommunityNode WiFi. Give them the password from the printed card or read it to them. They do NOT need the admin WiFi — CommunityNode has full access to the server.
2. Confirm ATAK is installed. ATAK-CIV (Android/Play Store) or iTAK (iPhone/App Store). If they don't have it, point them at the community page QR codes (<http://192.168.8.10:8081>) or hand them the field card.
3. Generate their enrollment QR code. Open OTS Web UI at <http://192.168.8.20:8080>. Log in with admin credentials. Navigate to Certificates (left sidebar) → Generate Data Package → enter callsign → Generate → click Show QR Code on the generated package.
4. User scans the QR code. In ATAK: Menu (☰) → Import → QR Code. Scan the QR from your screen. ATAK auto-imports server connection, certificate, and truststore. Connects within seconds.
5. User sets callsign. Settings → My Preferences → Callsign. Use their name or an assigned tactical callsign.
6. Verify. Their icon appears on the shared map. Test GeoChat: have them send a message. If using Mumble: open Mumla, server 192.168.8.20, port 64738.

## Fallback — Manual Server Entry

If the QR scan fails (old phone, camera issues):

ATAK → Settings → Network → TAK Servers → Add

| Setting  | Value        |
|----------|--------------|
| Server   | 192.168.8.20 |
| Port     | 8089         |
| Protocol | SSL          |

| Setting                       | Value             |
|-------------------------------|-------------------|
| Enable authentication         | Yes               |
| Enroll for client certificate | Yes               |
| Use default SSL/TLS certs     | No — uncheck this |
| Truststore password           | atakatak          |

ATAK contacts the server on port 8446, downloads and installs the certificate automatically. OTS handles all PKI — it auto-generates a unique client certificate for each enrolling device.

Time to onboard: 2–5 min with QR code. 5–10 min manual.

## Path B — Clearnet Browser (Remote Viewers)

Who: Remote team members, partner agencies, anyone with internet who needs situational awareness but does NOT need to be a CoT participant on the map.

What they get: Full OTS web map in a browser — all team positions, markers, missions, live updates. View and monitor. No app install required.

*⚠ Admin action required. Path B access is controlled by Cloudflare Zero Trust. Adding a user requires an admin with Cloudflare dashboard access and an internet connection. This cannot be done from the node itself. If possible, pre-add expected remote viewer emails before the event.*

### Operator Walkthrough

1. Contact the node admin (or perform if you have Cloudflare access). The admin must log in to dash.cloudflare.com from any internet-connected computer.
2. Add the email to the approved list: Zero Trust → Access → Applications → OTS Web Map → Edit Policy → Include rule (Email) → add address → Save. Access is granted immediately.
3. Send them the link and login instructions:

Web map: <https://tak.capableenough.org>

To log in:

1. Open the link in any browser
2. Enter your email address
3. Check inbox for a 6-digit PIN
4. Enter the PIN -- you're in  
(PIN expires in 10 minutes -- request a new one if needed)

4. Revoke access when needed. Zero Trust → Access → Applications → Edit Policy → remove their email → Save. Access is cut immediately.

### Automating Path B (Optional)

Adding emails one at a time through the dashboard is slow on site. A helper script using the Cloudflare API can automate this. If your admin has set up cf-access.sh on the comms Pi, the entire grant/revoke cycle becomes a single SSH command:

```
# Grant access (run from admin laptop on NodeAdmin WiFi):
ssh ps@192.168.8.10 sudo /usr/local/bin/cf-access.sh add user@example.com

# Revoke access:
ssh ps@192.168.8.10 sudo /usr/local/bin/cf-access.sh remove user@example.com
```

Ask your node admin to set this up before CCC26. Script requires a Cloudflare API token stored on the comms Pi.

Time to onboard: Under 60 seconds (add email, send link).

### Limitations

- They can see the map but do NOT appear as a position on it.
- No GeoChat, no marker creation, no mission participation from the browser.
- Requires internet access (the tunnel routes through Cloudflare).
- If you don't add their email, they can't get past the login screen.

## Path C — VPN + Full ATAK (Controlled Remote Access)

Who: Specific, trusted operators who need full ATAK participation from outside the node WiFi. This is NOT for general users. The operator personally sets this up for each individual.

What they get: Full ATAK over VPN — identical to being on local WiFi. Live position sharing, GeoChat, markers, missions, everything. They appear on the map as a full team member.

Remote user requirements: Tailscale (Play Store) + ATAK-CIV (Play Store).

### Operator Steps

Requires: Admin machine connected to CommunityNode WiFi or NodeAdmin WiFi. Run SSH commands from your admin laptop — not from the node itself.

1. Generate a VPN auth key (on tactical Pi):

```
ssh ps@192.168.8.20
sudo docker exec tactical-node-headscale-1 \
  headscale preauthkeys create --user community --expiration 24h
# For standing team member (persistent):
sudo docker exec tactical-node-headscale-1 \
  headscale preauthkeys create --user community --expiration 90d --reusable
```

2. Send the user these instructions:

1. Install Tailscale from the Play Store
2. Open Tailscale > tap profile icon > Accounts > three-dot menu  
> 'Use an alternate server'
3. Enter: `https://m2vpn.capableenough.org`
4. Paste the auth key I sent you
5. Tailscale connects -- you now have a VPN tunnel to the node
6. Open ATAK > Settings > Network > TAK Servers > Add  
Server: 192.168.8.20 | Port: 8089 | Protocol: SSL  
Check: Enroll for Client Certificate  
Uncheck: Use default SSL/TLS Certificates  
Truststore password: atakatak
7. Set callsign: Settings > My Preferences > Callsign
8. You should appear on the shared map within seconds.

3. Verify their connection:

```
ssh ps@192.168.8.20
sudo docker exec tactical-node-headscale-1 headscale nodes list
# Confirm their device has an assigned IP (100.64.x.x range)
```

4. Revoke access when done:

```
ssh ps@192.168.8.20
sudo docker exec tactical-node-headscale-1 headscale nodes delete --identifier <id>
# User's Tailscale immediately disconnects. No residual access.
```

### Security Notes

- Each auth key is tied to an expiration — it cannot be reused after it expires.
- You control exactly who has access and can revoke instantly.
- VPN traffic is encrypted end-to-end (WireGuard).
- The Headscale server only accepts connections with valid pre-auth keys.
- Subnet routes (192.168.8.0/24) are already approved on both Pis — remote users reach LAN services as if local.

## During the Event

### Monitoring

- Kiosk touchscreen shows live service status. Green = healthy.
- OTS web map at `http://192.168.8.20:8080` shows all connected EUDs.

### Positions Stop Updating

The CoT parser service processes incoming position reports. If positions freeze, restart it:

```
ssh ps@192.168.8.20
systemctl status cot_parser
sudo systemctl restart cot_parser
```

## User Can't Connect to ATAK

- Wrong WiFi. Must be on CommunityNode, not NodeAdmin or a hotspot.
- SSL checkbox. In ATAK TAK Servers, must have 'Enroll for Client Certificate' checked and 'Use default SSL/TLS Certificates' unchecked.
- Old server entry. Delete the server entry in ATAK and re-add from scratch. Stale certificates cause repeated connection failures.
- Private DNS / VPN on phone. Disable any system-level VPN or Private DNS setting before connecting. These interfere with local resolution.
- Certificate enrollment already done. If the user previously enrolled, ATAK may have cached a bad cert. Delete the server and re-add.

## Element / Matrix Down

```
ssh ps@192.168.8.10
docker restart community-node-conduit-1
docker restart community-node-nginx-1
```

## User Not on Map

- In ATAK: Settings → Reporting → Reporting Strategy → set to Consistent.
- Or: toggle server off/on in TAK Servers to force a reconnect.
- Confirm GPS is active on their phone and location permission is granted to ATAK.
- Check OTS is showing their callsign in the Clients list (<http://192.168.8.20:8080>).

# Teardown

Shut down in reverse power-on order:

1. Notify all active ATAK users: 'Server going offline in 2 minutes.'
2. Export any mission data from OTS Web UI if needed (Missions → Export).
3. SSH to Pi #1: `sudo shutdown -h now`
4. SSH to Pi #2: `sudo shutdown -h now`
5. Wait 30 seconds for both Pis to power off (LEDs go dark).
6. Power off the GL.iNet router.
7. Power off the TP-Link switch.
8. Power off the Anker charger.
9. Power off the Tupavco PDU.
10. UPS last: hold power button until it beeps off.

## Pack-Up Notes

- Disconnect LoRa antennas before coiling USB cables — the SMA connectors are fragile.
- USB-C cables for the Pis go in the accessory pouch, not loose in the rack.
- Monero blockchain state is on NVMe — no action needed; it persists across reboots.
- Tor hidden service keys are in `/var/lib/tor/` — backed up with the full config backup.



## Emergency: Internet Lost

All local services continue. ATAK, voice, mesh, and Element all work with no internet. The Cloudflare Tunnel and Headscale VPN will drop — remote viewers lose access. Everyone on the local WiFi is unaffected.

To air-gap intentionally (stop all cleartext egress):

```
ssh ps@192.168.8.10
docker stop community-node-cloudflared-1

ssh ps@192.168.8.20
docker stop tactical-node-headscale-1
```

## Emergency: Pi Won't Boot

Diagnostic steps:

1. Confirm USB-C power cable is seated (Pis use USB-C PD — not all cables work).
2. Check Anker charger output LED — should be solid, not blinking.
3. Connect a monitor via micro-HDMI to see boot output.
4. If kernel panic / filesystem error: boot from microSD with rescue image.
5. If NVMe not found: check M.2 seating in the GeekPi bracket.

### What Still Works If a Pi Goes Down

| Pi Down          | Services Lost                                               | Services Still Available                                          |
|------------------|-------------------------------------------------------------|-------------------------------------------------------------------|
| Pi #1 (comms)    | Matrix/Element, Tor, I2P, Cloudflare Tunnel, AdGuard DNS    | ATAK (Pi #2), Mumble, Meshtastic, Reticulum, Monerod, Headscale   |
| Pi #2 (tactical) | ATAK/OTS, Mumble, Meshtastic, Reticulum, Monerod, Headscale | Matrix/Element, community kiosk, Tor hidden services, AdGuard DNS |

# ATAK Configuration

## What's Built In — No Plugins Needed

- Position sharing (PLI) — live team positions on shared map
- GeoChat — group and private messaging within ATAK
- Markers and drawing tools
- Route planning
- CASEVAC reports (9-line)
- ExCheck checklists
- Track logging
- Offline maps (with pre-cached tiles served by OTS)
- Terrain analysis
- Geofencing

## Recommended Plugins (2 Only)

| Plugin                          | Purpose                                    | Install                    | Who Needs It                                          |
|---------------------------------|--------------------------------------------|----------------------------|-------------------------------------------------------|
| Meshtastic ATAK Plugin          | Direct radio-to-phone bridge for LoRa mesh | <i>GitHub APK sideload</i> | <i>Operators with their own Meshtastic radio only</i> |
| VNS (Vehicle Navigation System) | In-ATAK turn-by-turn navigation            | <i>Google Play Store</i>   | <i>Team leads running vehicle ops</i>                 |

## Data Packages Pre-Loaded

| Package               | Contents                  | Auto-push                          |
|-----------------------|---------------------------|------------------------------------|
| DTED2 Florida         | 30m elevation data        | <i>Via enrollment data package</i> |
| PMTiles FL (578MB)    | Offline vector maps z0-14 | <i>Served at 192.168.8.20:8080</i> |
| Community Node marker | M2 HQ position            | <i>Cron push every 4 hours</i>     |

*Offline map tiles are already cached — no internet needed for map display.*

## Matrix / Element Chat

*M2 Conduit server — standalone field node. Independent of clearnet Matrix.*

## Access

| Method             | URL                                                                               |
|--------------------|-----------------------------------------------------------------------------------|
| On node WiFi       | <a href="http://192.168.8.10:8080">http://192.168.8.10:8080</a>                   |
| Clearnet           | <a href="https://element.capableenough.org">https://element.capableenough.org</a> |
| Matrix server name | <a href="https://m2.capableenough.org">m2.capableenough.org</a>                   |

## Room Structure

Community Space — standing rooms:

| Room         | Purpose                                  |
|--------------|------------------------------------------|
| #comms       | Radio, mesh, Reticulum, Meshtastic       |
| #medical     | First aid, trauma, community health      |
| #agriculture | Food production, permaculture            |
| #security    | Physical security, situational awareness |
| #logistics   | Supply chain, transportation             |
| #general     | Social, announcements                    |
| #onboarding  | Encryption setup help, welcome resources |

*Dynamic rooms are spun up as needed (mission-specific, event, emergency).*

## Encryption

- All rooms encrypted by default (E2EE enforced).
- Once encryption is on, it cannot be disabled (protocol enforced).
- New users: set up key backup before joining rooms to prevent message loss.

## Roles

| Role      | Power Level | Who                                    |
|-----------|-------------|----------------------------------------|
| Admin     | 100         | Operator + trusted family (1–2 people) |
| Moderator | 50          | Promoted community members             |
| Member    | 0           | Standard users                         |

## Quick Reference — URLs & Ports

| Service                | LAN URL                                                                       | Port  | Notes                                                                                 |
|------------------------|-------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------|
| Router admin           | http://192.168.8.1                                                            | 80    | <i>GL.iNet admin panel</i>                                                            |
| Community page         | http://192.168.8.10:8081                                                      | 8081  | <i>Operator onboarding kiosk</i>                                                      |
| Element Web            | http://192.168.8.10:8080                                                      | 8080  | <i>Also: <a href="http://element.capableenough.org">element.capableenough.org</a></i> |
| OTS web map (LAN)      | http://192.168.8.20:8080                                                      | 8080  | <i>Admin login required</i>                                                           |
| OTS web map (clearnet) | <a href="http://tak.capableenough.org">tak.capableenough.org</a>              | 443   | <i>Cloudflare Access — OTP email login</i>                                            |
| ATAK server (SSL)      | 192.168.8.20                                                                  | 8089  | <i>After cert enrollment</i>                                                          |
| ATAK enrollment        | 192.168.8.20                                                                  | 8446  | <i>Auto-used during first connect</i>                                                 |
| Mumble voice           | 192.168.8.20                                                                  | 64738 | <i>No password required by default</i>                                                |
| AdGuard Home           | http://192.168.8.10:3000                                                      | 3000  | <i>DNS admin — <code>admin/admin</code> default</i>                                   |
| Matrix federation      | http://192.168.8.10:8448                                                      | 8448  | <i>Conduit federation port</i>                                                        |
| Headscale admin        | <a href="https://m2vpn.capableenough.org">https://m2vpn.capableenough.org</a> | 443   | <i>Via Cloudflare Tunnel</i>                                                          |
| Monerod RPC            | http://192.168.8.20:18081                                                     | 18081 | <i>Local RPC only — not exposed to WAN</i>                                            |

## WiFi Networks

| SSID          | Purpose                   | Access                                  |
|---------------|---------------------------|-----------------------------------------|
| CommunityNode | Field operator / EUD WiFi | Full node access (all services)         |
| NodeAdmin     | Tech admin only           | Router admin access + all node services |

## Default Credentials

Change all passwords before deployment. Sensitive values are in M2\_SECRETS.md (gitignored — local only).

| Service                | Username                    | Password            |
|------------------------|-----------------------------|---------------------|
| ATAK enrollment        | field                       | community           |
| OTS admin              | admin                       | [see M2_SECRETS.md] |
| GL.iNet admin          | root                        | [see M2_SECRETS.md] |
| AdGuard Home           | admin                       | [see M2_SECRETS.md] |
| Matrix admin           | @admin:m2.capableenough.org | [see M2_SECRETS.md] |
| Mumble SuperUser       | (blank)                     | [see M2_SECRETS.md] |
| ATAK truststore (cert) | (n/a)                       | atakatak            |